

2020

Advanced Computing Program

Entrance Exam

January 6, 2020

Problem S1: Cutting String

Time limit: 1.0 second(s)

Memory limit: 512 MiB

Description

We define a string to be *pretty*, if the string is non-empty and all of its characters are the same. For example, strings 6666, aaa and 9 are *pretty*, while strings 2020, aaabaaa are not.

Given a string as input, you should cut the string into multiple parts, and each part must be *pretty*. You should minimize the number of parts and output a possible solution.

You may want to see the samples below for better understanding.

Formally, each part must be a consecutive substring of the input, and the concatenation of all parts must be the input string.

Input Format

Read from standard input.

The only line of input is a string. The string is non-empty and only consists of digits (0 to 9) and lower-case letters (a to z).

Note that a newline character (\n) will be present at the end of the line. This character does not belong to the input string.

Output Format

Write to the standard output.

The first line is an integer m , denoting the minimal number of parts.

For next m lines, you should output each part in order, one per line.

Sample Input 1

```
6666aaa9
```

Sample Output 1

```
3
6666
```

aaa

9

Sample Input 2

2020aaabaaa

Sample Output 2

7

2

0

2

0

aaa

b

aaa

Sample Input 3

loooooongstrrrrrring

Sample Output 3

10

l

oooooo

n

g

s

t

rrrrr

i

n

g

Subtasks

Let n denote the length of input string.

Subtask 1 (19 points)

Cases in this subtask satisfy $n = 1$.

Subtask 2 (18 points)

Cases in this subtask satisfy $n \leq 2$.

Subtask 3 (14 points)

Cases in this subtask satisfy $n \leq 100$, and $m = 1$ for the correct output.

Subtask 4 (21 points)

$n \leq 100, m \leq 2$.

Subtask 5 (19 points)

$n \leq 100$.

Subtask 6 (9 points)

$n \leq 100000$.

Problem S2: Super Shorthand

Time limit: 1.0 second(s)

Memory limit: 512 MiB

Background

We have defined *pretty* strings in the previous task.

Description

We define the *shorthand* string of a *pretty* string, to be the length of the *pretty* string (in string form), followed by the first character of the *pretty* string. For example, the *shorthand* of aaa is 3a, because aaa is three a-s. The *shorthand* of 6666 is 46, and the *shorthand* of gggggggggggg is 11g.

If a string is not *pretty*, we define its *shorthand* as follow. First we cut the string into minimal parts of *pretty* strings (as you did in the previous problem). Then we find the *shorthands* of those *pretty* strings. Finally we concatenate the *shorthands* in order, and we get the *shorthand* of the original string. For example, the *shorthand* of aaa6666gggggggggggg is 3a4611g, and the *shorthand* of abcd is 1a1b1c1d.

Given a string and a positive integer k as input, you should perform k times of *shorthand* operation on string, and output the result.

Input Format

Read from standard input.

The first line of input is an integer k .

The second line of input is a string. The string is non-empty and only consists of digits (0 to 9) and lower-case letters (a to z).

Note that a newline character (\n) will be present at the end of the line. This character does not belong to the input string.

Output Format

Write to the standard output.

The only line of output is a string, denotes the result.

However, if the full result is strictly longer than 100000 characters, you should only output the first 100000 characters, and then output You can refer to the last

sample for more details.

Sample Input 1

```
1
gggggggggggg2020abbccdddd
```

Sample Output 1

```
11g121012101a2b3c4d
```

Sample Input 2

```
2
gggggggggggg2020abbccdddd
```

Sample Output 2

```
211g1112111011121110111a121b131c141d
```

Sample 3

See *3.in* and *3.ans* in the problem path.

Subtasks

Let n denote the length of input string, and L denote the length of the full result.

Subtask 1 (13 points)

$$n \leq 100, k = 1.$$

Subtask 2 (17 points)

$$n \leq 100, k \leq 10.$$

Subtask 3 (16 points)

$$n \leq 100000, L \leq 100000, n \leq 1000.$$

Subtask 4 (23 points)

$$n = 1, k \leq 1000.$$

Subtask 5 (12 points)

$$n \leq 2, k \leq 1000.$$

Subtask 6 (19 points)

$$n \leq 100000, k \leq 1000.$$

Problem S3: Solve the Ones

Time limit: 1.0 second(s)

Memory limit: 512 MiB

Description

We define the number $A(m, k)$ to be k digits of 1 of base- m . For example, $A(10, 6) = 111111_{(10)}$, $A(3, 5) = 11111_{(3)} = 121_{(10)}$.

Given two positive integers n and m , you need to output the minimal positive integer k such that $A(m, k)$ is a multiple of n . You also need to judge the case that no solution k exists.

Input Format

Read from standard input.

The only line of input consists of two integers n and m , separated by a space.

Output Format

Write to the standard output.

The only line of output contains a single integer, denoting the minimal positive k . If such k does not exist, you should output 0 instead.

Sample Input 1

31 2

Sample Output 1

5

Sample Input 2

31 10

Sample Output 2

15

Sample Input 3

20 10

Sample Output 3

0

Subtasks

All subtasks satisfy $n \geq 2$ and $2 \leq m \leq 16$.

Subtask 1 (11 points)

$n \leq 5$. We guarantee the required k exists.

Subtask 2 (12 points)

$n \leq 5$.

Subtask 3 (13 points)

$n \leq 15$.

Subtask 4 (17 points)

$n \leq 10^6$.

Subtask 5 (30 points)

$n \leq 10^{12}$ and n is a prime number.

Subtask 6 (17 points)

$n \leq 10^{12}$.